

Load Interconnection Request System Impact Study Report

For Queue Project Q23-2307

Lone Tree 138 kV
July 19, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2307. The Interconnection Customer proposes a 452.7 MW large load (data center) facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on April 23, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 603 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,587,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LOWMOOR 345/138 kV Transformer Bank	ASPEN - LOWMOOR 345.0 kV	107.3	586	41.6
BLUEBONNET - JONES CHAPEL 230.0 kV	ROBISON PARK - RUSSELL FORK 230.0 kV	107.1	1,701	6.1
RED HILLS - VASEAUX LAKE 138.0 kV	LOWMOOR - HINTON AP 345.0 kV	102.5	507	7.0
SLOCUM RD SUBSTATION - MCNEIL 230.0 kV	RICHLAND - DEKALB 500.0 kV	102.4	2,550	5.4
MCCLAIN ENERGY FACILITY - VASEAUX LAKE 138.0 kV	HINTON AP - LOWMOOR 345.0 kV	102.4	511	7.0
SLOCUM RD - DEKALB 230.0 kV L	RICHLAND - DEKALB 500.0 kV	102.4	2,585	5.4
BREED - MCNEIL 230.0 kV	DEKALB - RICHLAND SUBSTATION 500.0 kV	102.4	2,568	5.4
BEARDSLEY ST - RED HILLS 138.0 kV	LOWMOOR - HINTON AP 345.0 kV	102.1	496	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOWMOOR 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$18,830,000	19% (\$3,517,000)
JONES CHAPEL SUBSTATION - BLUEBONNT 230.0 kV	Reconductor line	\$17,437,000	3% (\$548,000)
RED HILLS - VASEAUX LAKE 138.0 kV	Reconductor line	\$5,949,000	4% (\$262,000)
SLOCUM RD SUBSTATION - MCNEIL 230.0 kV	Reconductor line	\$5,205,000	3% (\$174,000)
MCCLAIN ENERGY FACILITY - VASEAUX LAKE 138.0 kV line	Reconductor line	\$3,568,000	4% (\$157,000)
DEKALB SUBSTATION - SLOCUM RD 230.0 kV line	Reconductor line	\$15,081,000	3% (\$503,000)
BREED - MCNEIL 230.0 kV	Reconductor line	\$7,356,000	3% (\$243,000)
BEARDSLEY ST - RED HILLS 138.0 kV line	Reconductor line	\$4,156,000	4% (\$183,000)
	Total allocated to project		\$5,587,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.